OJT Record Management Application: Progress Documentation

Project: ***On-the-Job Training (OJT) Record Management Application***

Group: *3*

Repository: [swvenz/Project-OJT-RMA-System]

Member	Role
Russell Shaun Simon	Project Manager
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Jared Ramos	Quality Assurance
Rovi Dela Rosa	Backend DEV
Andrei Sietereales	Tech Doc
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System overview

The system replaces paper files and separate spreadsheets with one PostgreSQL database for OJT records. It stores student profiles, OJT applications, requirements, company placements, attendance, training hours and evaluations, so records can be searched, updated and kept consistent. This implements the blueprint approved in Activity 2.

Scope. Student and company records, application and requirement tracking, placement records, attendance monitoring, evaluation records, OJT status and record viewing.

Out of scope. Recruitment and job matching, salary and financial transactions, and built-in messaging.

The main functions:

#	Function	Priority
1	Register student information	High
2	Register company information	High
3	Track student attendance	High
4	Evaluate student	High
5	Track OJT placement	High
6	Track OJT requirements	Medium
7	Monitor OJT status	Medium
8	Help students with OJT application	Low

User roles

The application has five user roles plus one system account, each limited to the functions below. Application roles are stored in `users.role`; the runtime account is the only login the application uses to reach PostgreSQL.

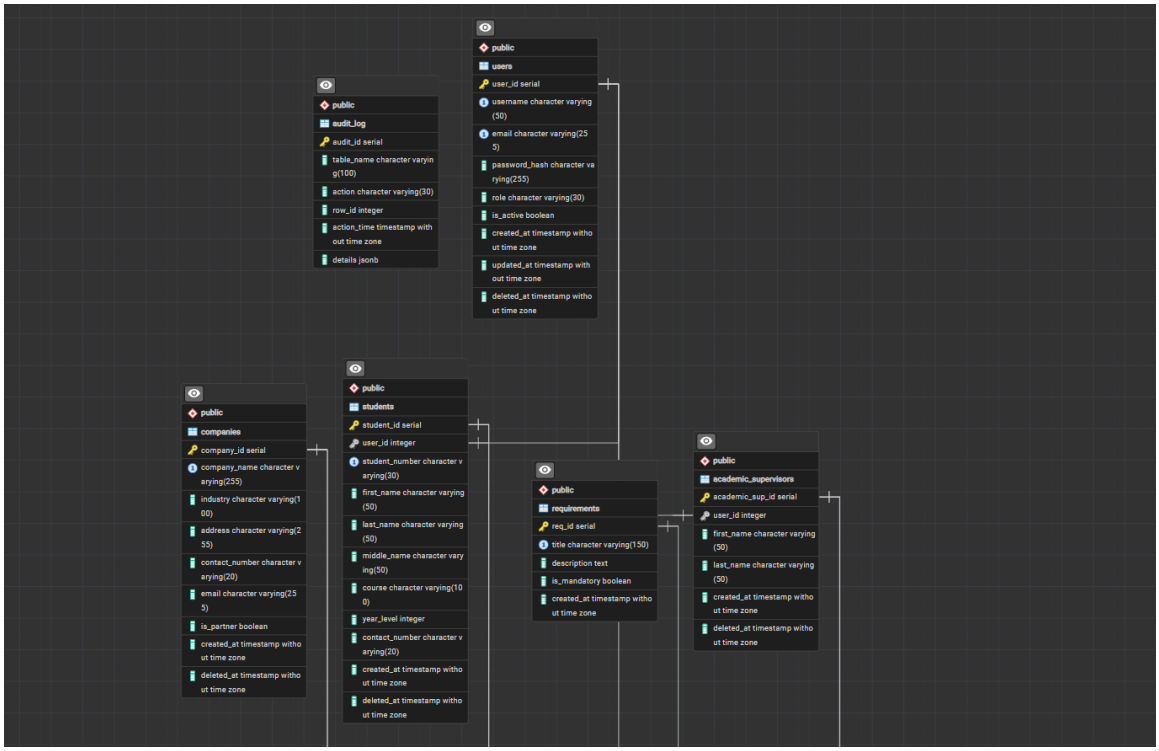
Application role	Responsibilities	Functions and data access	DB role / account
Administrator (`ADMIN`)	Manages accounts and views the audit log.	Create an account (user and profile, all-or-nothing). Activate or deactivate accounts, but not their own.	*ojt_admin*
Student (`STUDENT`)	Applies for OJT, submits requirements, logs attendance.	Read partner companies. Create one active application at a time. Create or update own requirement submissions. Create attendance for own active placement. Read own placement and evaluations.	*ojt_student*
Academic supervisor (`ACADEMIC_SUPERVISOR`)	Verifies requirements and attendance; closes placements.	Verify or reject pending requirements. Verify attendance of own placements. Mark own placement Completed. Read placements and evaluations.	*ojt_academic_supervisor*
Company supervisor (`COMPANY_SUPERVISOR`)	Supervises interns on site and evaluates them.	Read own interns. Verify attendance of own interns. Create evaluations for own placements.	*ojt_company_supervisor*
OJT coordinator (`OJT_COORDINATOR`)	Reviews applications; manages	Approve an application (creates the placement, assigns supervisors, dates and hours) or reject it with a reason.	*ojt_coordinator*

Application role	Responsibilities	Functions and data access	DB role / account
	companies and placements.	Add companies. Read all placements.	

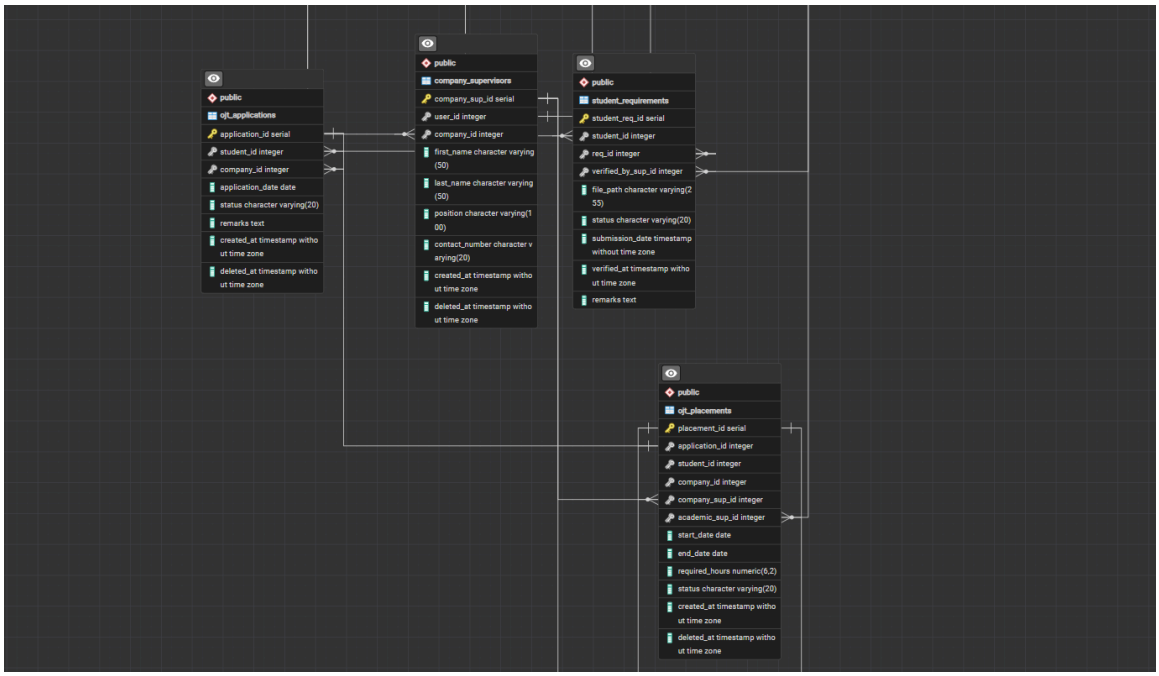
Database implementation

The PostgreSQL schema has 12 tables in the `public` schema, created by `database/OJT Database.sql` in the repository. The ERD below is exported from pgAdmin and matches the implemented schema.

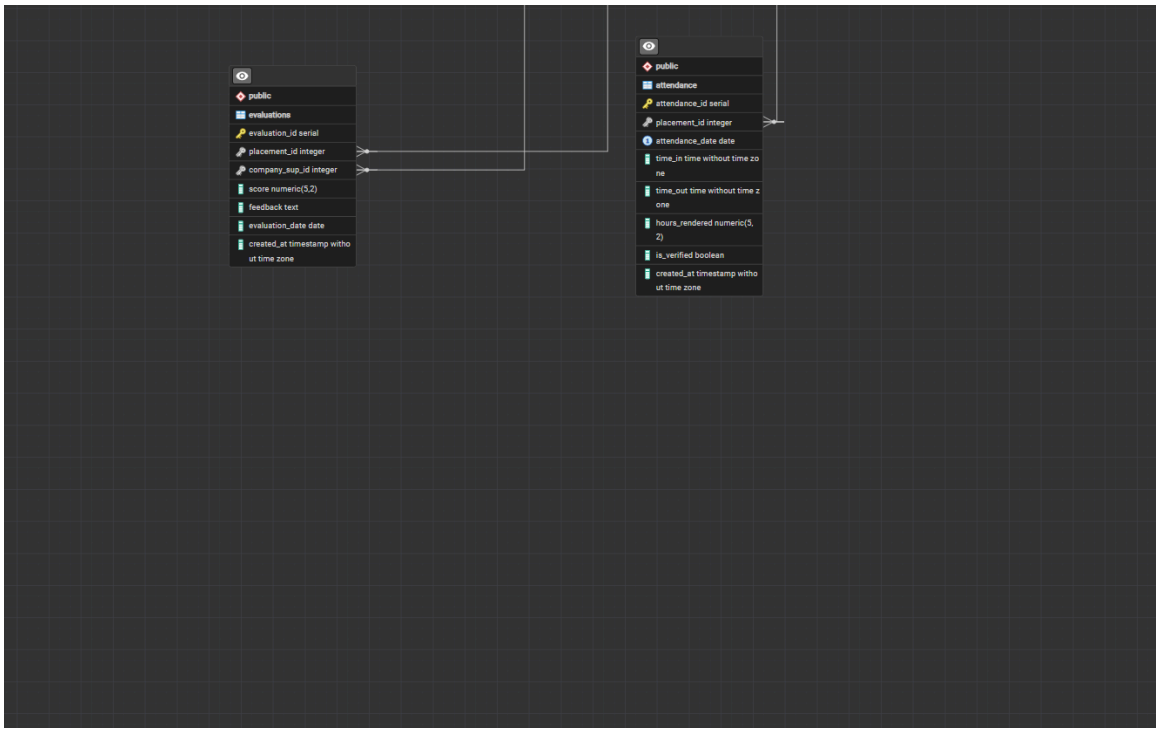
ERD (pgAdmin)



Part 1: audit_log, users, companies, students, requirements, academic_supervisors.



Part 2: ojt_applications, company_supervisors, student_requirements, ojt_placements.



Part 3: evaluations, attendance.

Table	Primary key	Foreign keys	Key constraints
users	user_id	none	Unique username and email; role limited to 5 values; is_active defaults to true
audit_log	audit_id	none	details stored as JSONB
students	student_id	user_id → users	Unique user_id and student_number; year_level > 0
companies	company_id	none	Unique company_name; is_partner defaults to false
academic_supervisors	academic_sup_id	user_id → users	Unique user_id
company_supervisors	company_sup_id	user_id → users, company_id → companies	Unique user_id
requirements	req_id	none	Unique title; is_mandatory defaults to true
ojt_applications	application_id	student_id → students, company_id → companies	Status: Pending, Approved, Rejected, Withdrawn; unique (application_id, student_id, company_id)
student_requirements	student_req_id	student_id, req_id, verified_by_sup_id	Unique (student_id, req_id); status: Pending, Submitted, Approved, Rejected
ojt_placements	placement_id	application_id, student_id, company_id, company_sup_id, academic_sup_id	Unique application_id; required_hours > 0; start_date ≤ end_date; status: Pending,

Table	Primary key	Foreign keys	Key constraints
			Active, Completed, Terminated
attendance	attendance_id	placement_id → ojt_placements	Unique (placement_id, attendance_date); hours_rendered ≥ 0; time_out ≥ time_in
evaluations	evaluation_id	placement_id, company_sup_id	score between 0 and 100

Scripts and seed data

The repository holds the table-creation script, the five PostgreSQL login roles and their grants, and one placeholder account per application role. The placeholder password hash is not valid for login. Full seed data is to be added.

Design changes since Activity 2

Change	Reason
Added a composite unique key on `ojt_applications` and a matching composite foreign key from `ojt_placements`.	A placement's student and company must match the approved application it came from.
Added `UNIQUE (placement_id, attendance_date)` on `attendance`.	Enforces one attendance entry per placement per day (business rule 3).
Added `UNIQUE (student_id, req_id)` on `student_requirements`.	A student submits each requirement once.
Changed `attendance.time_in` and `time_out` from timestamp to time, and `hours_rendered` from numeric(4,2) to numeric(5,2).	The same date can be recorded for different placements, but duplicate attendance for the same placement and date is prevented.
Dropped `employee_number`, `department` and `contact_number` from the supervisor tables.	It was not needed anymore.
Added a `role` check limited to ADMIN, STUDENT, ACADEMIC_SUPERVISOR, COMPANY_SUPERVISOR and OJT_COORDINATOR.	Matches the five application roles in the user roles section.

Application evidence

The Java desktop app runs against the database and shows role-specific screens for the student, company supervisor and administrator.

Role	Screen	What the screenshot shows
All	Sign in	Username and password form with demo accounts listed
Student	Partner companies	One company (Acme Corp, Technology, Manila) with an Apply to selected company button
Student	My applications	Empty list, so no application has been created yet
Student	My requirements	Three mandatory requirements (Resume, Parent consent, Medical certificate), all Not submitted, with a Submit selected requirement button
Student	Attendance, My placement, My evaluations	Empty lists; Attendance has a Record attendance button
Company supervisor	My interns, Attendance to verify, Evaluations	Empty lists with Evaluate selected intern and Verify selected buttons
Administrator	Accounts	11 accounts across the roles, with Create account and Activate / deactivate buttons; account 11 (student role) was added after the demo accounts
Administrator	Audit log	Empty list

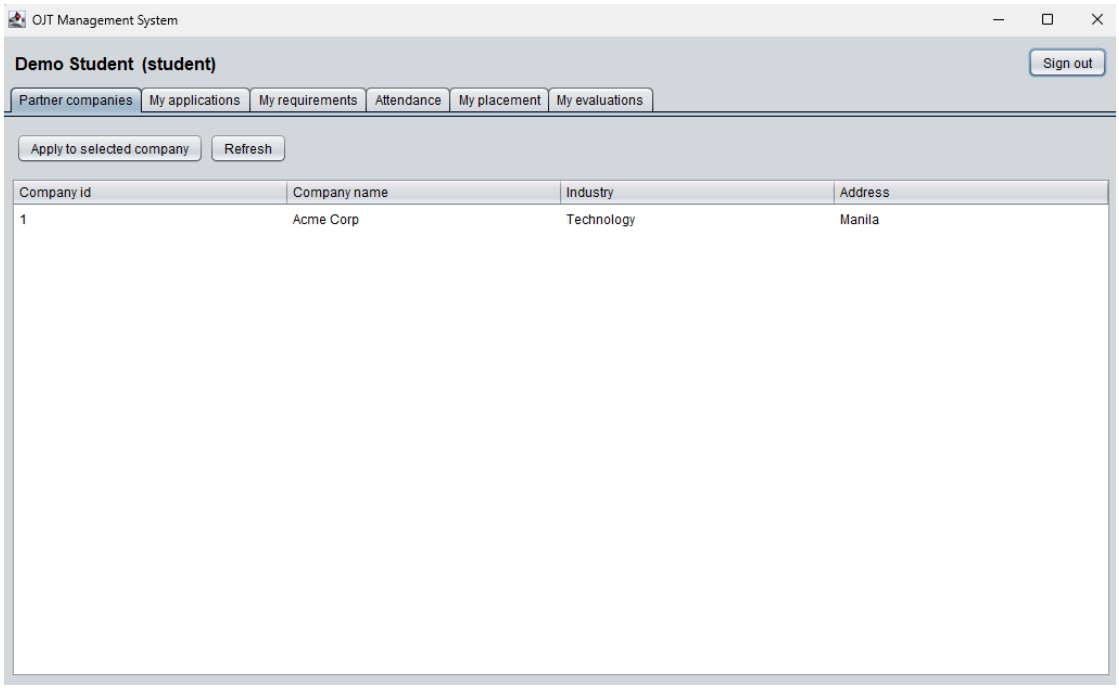
Screenshots

Sign in

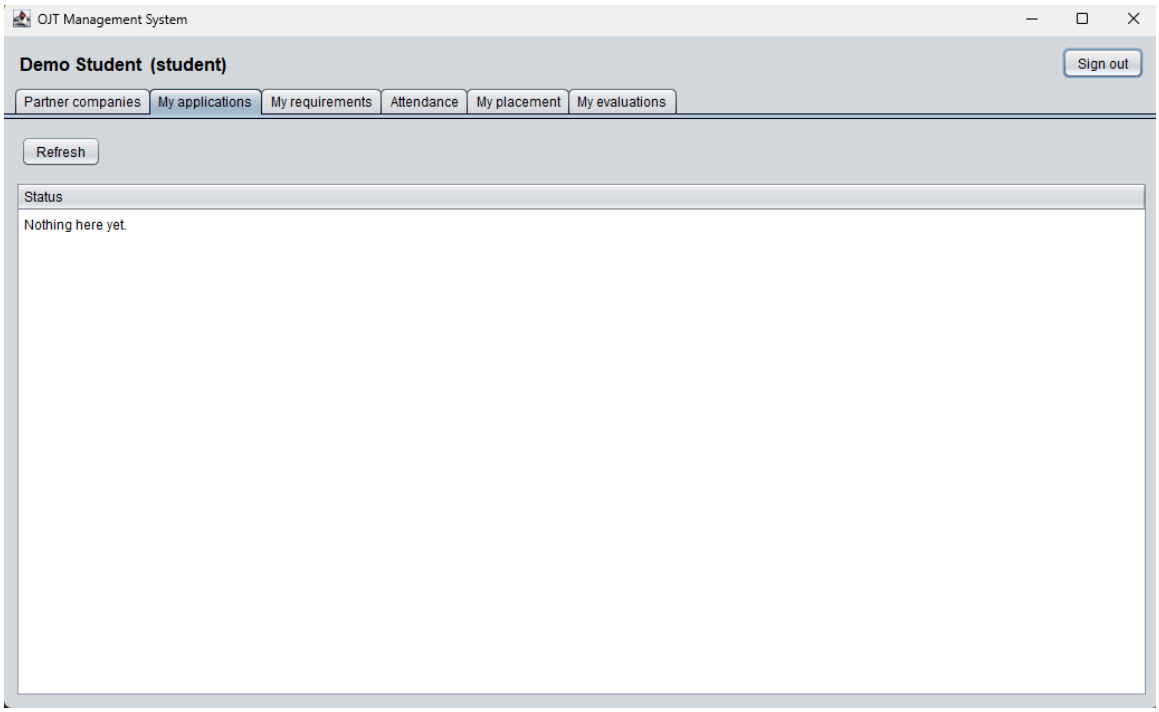


Sign-in screen (demo account details cropped out).

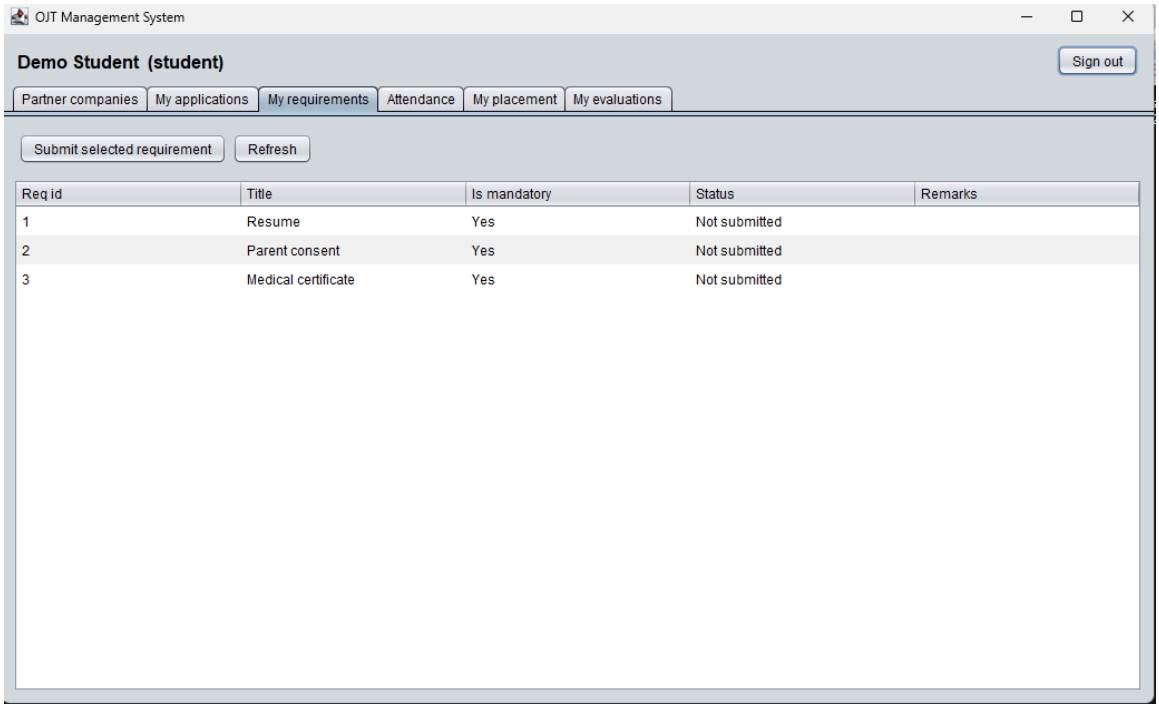
Student



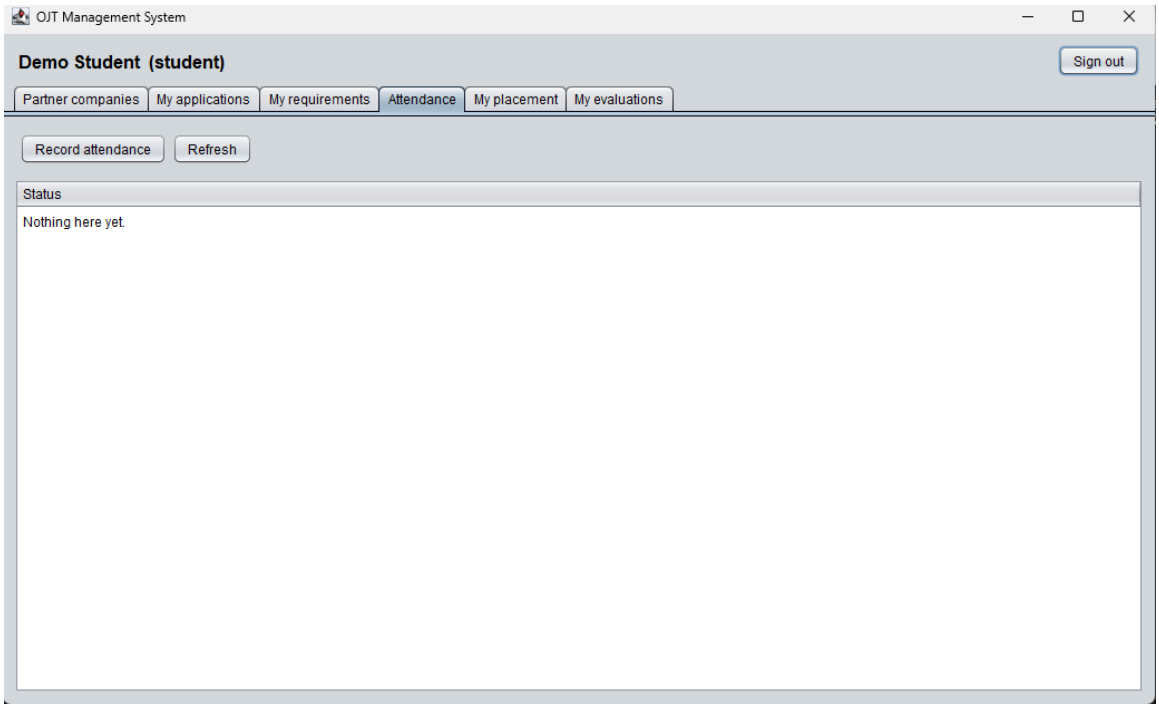
Student: Partner companies.



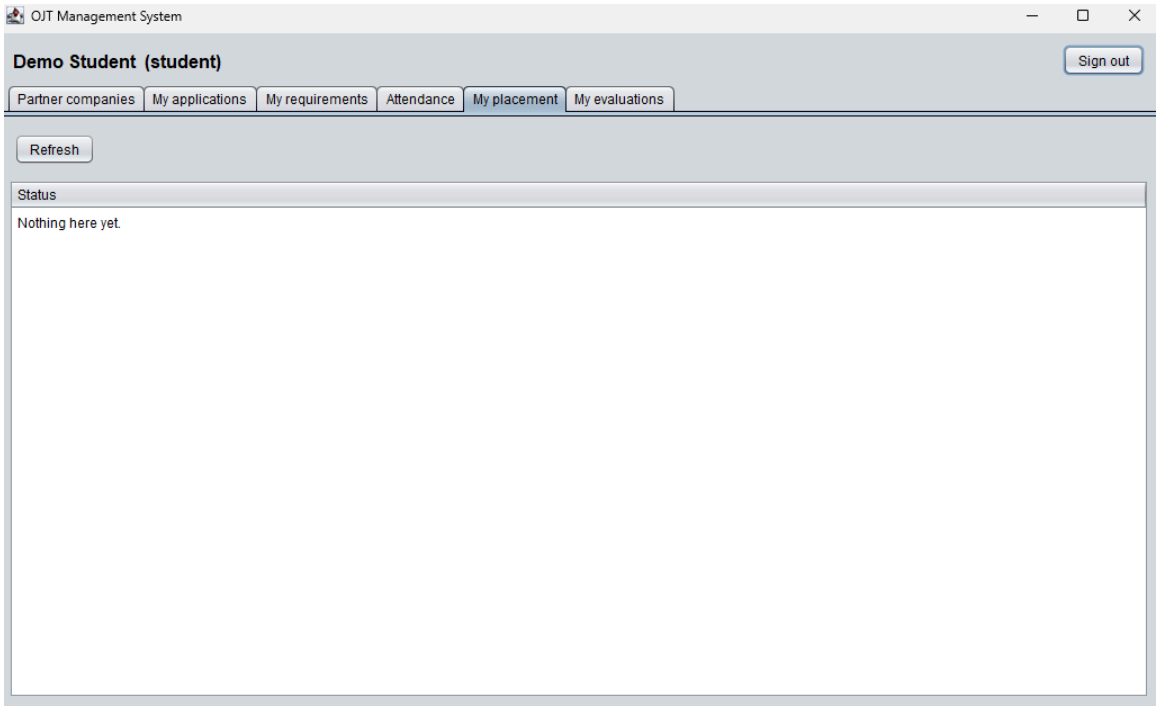
Student: My applications.



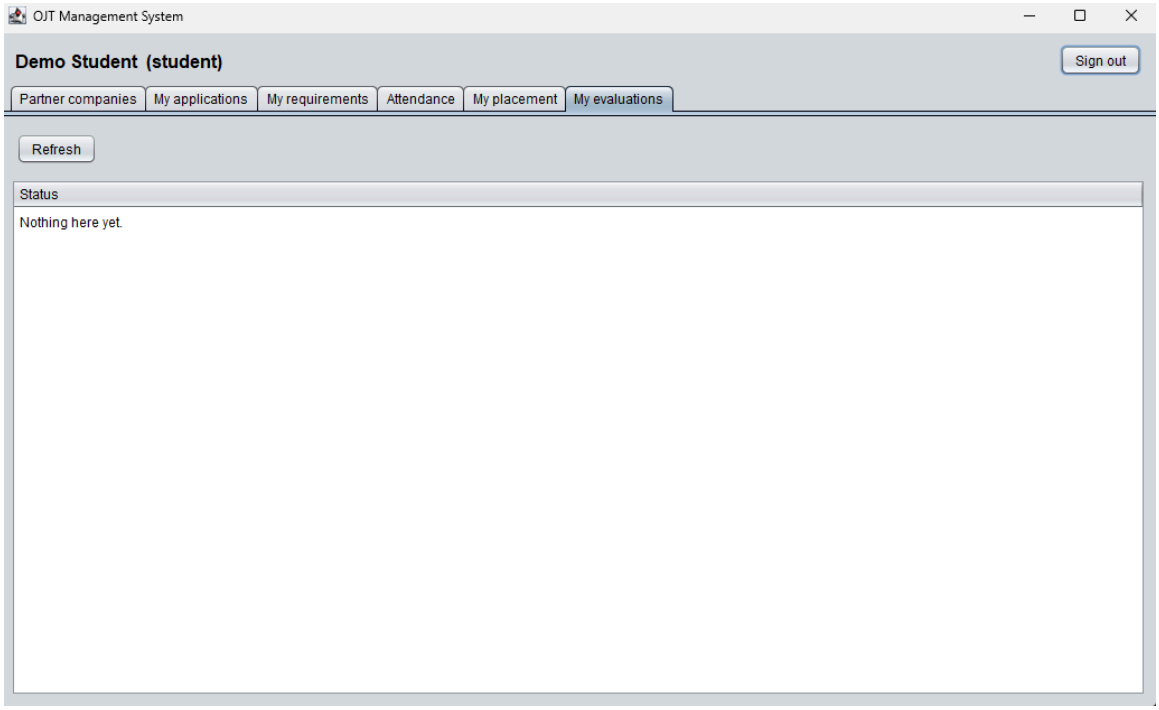
Student: My requirements.



Student: Attendance.

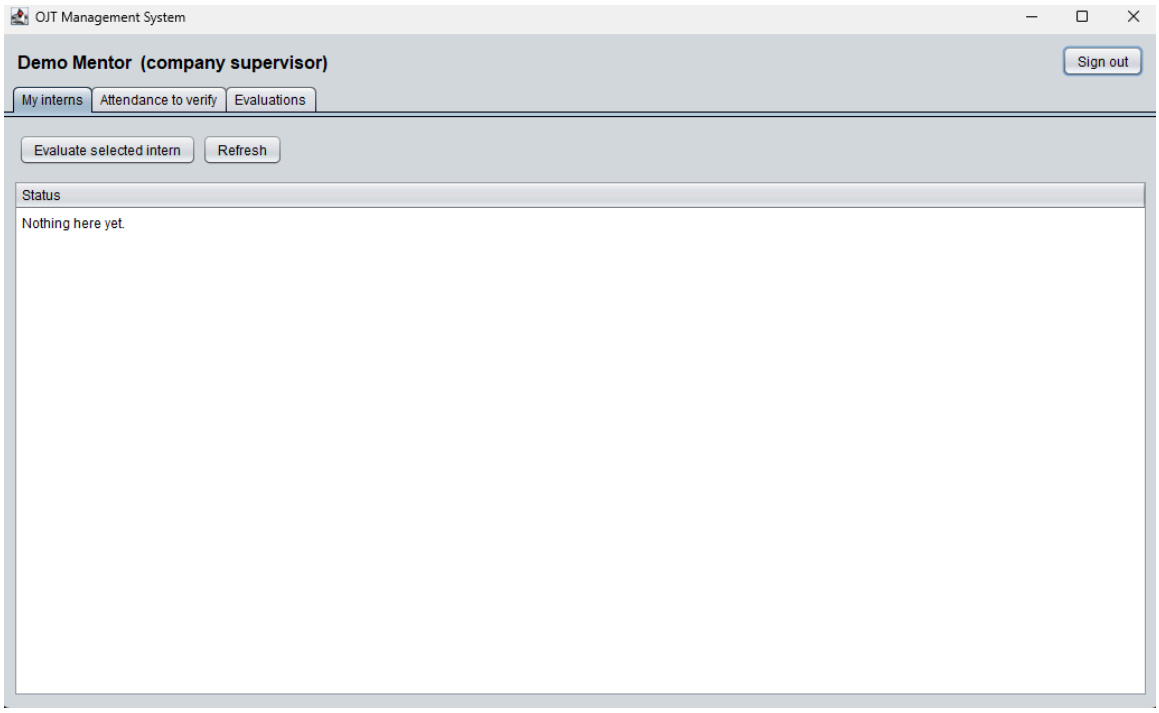


Student: My placement.

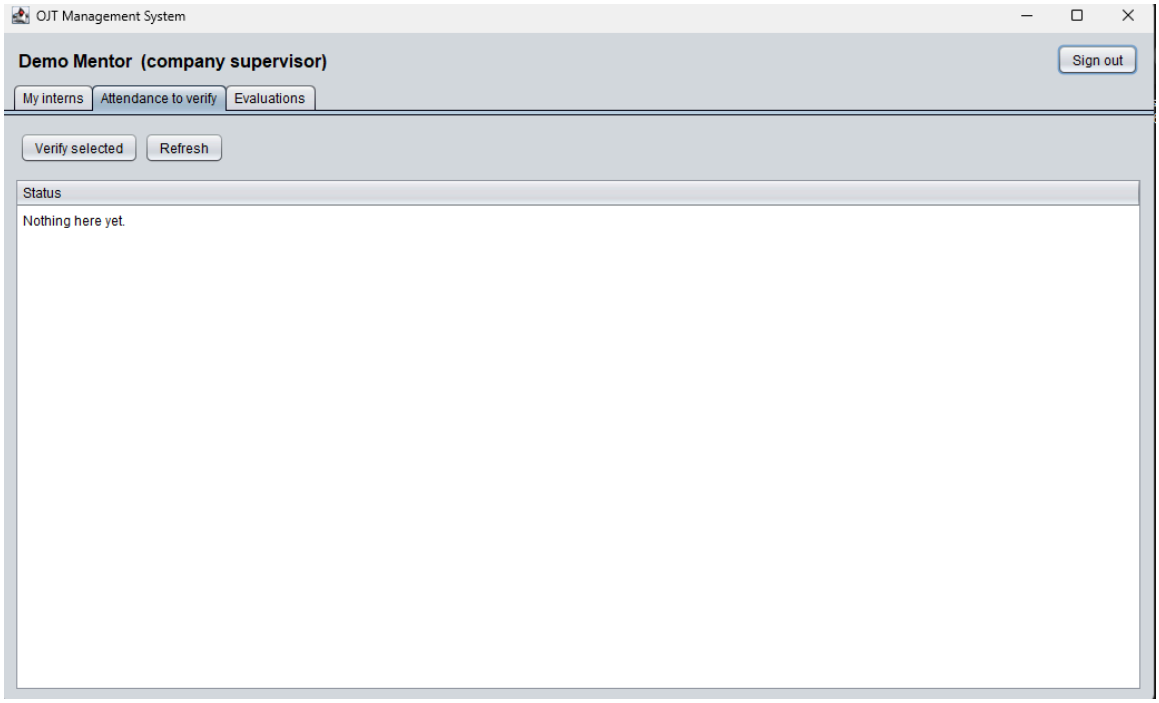


Student: My evaluations.

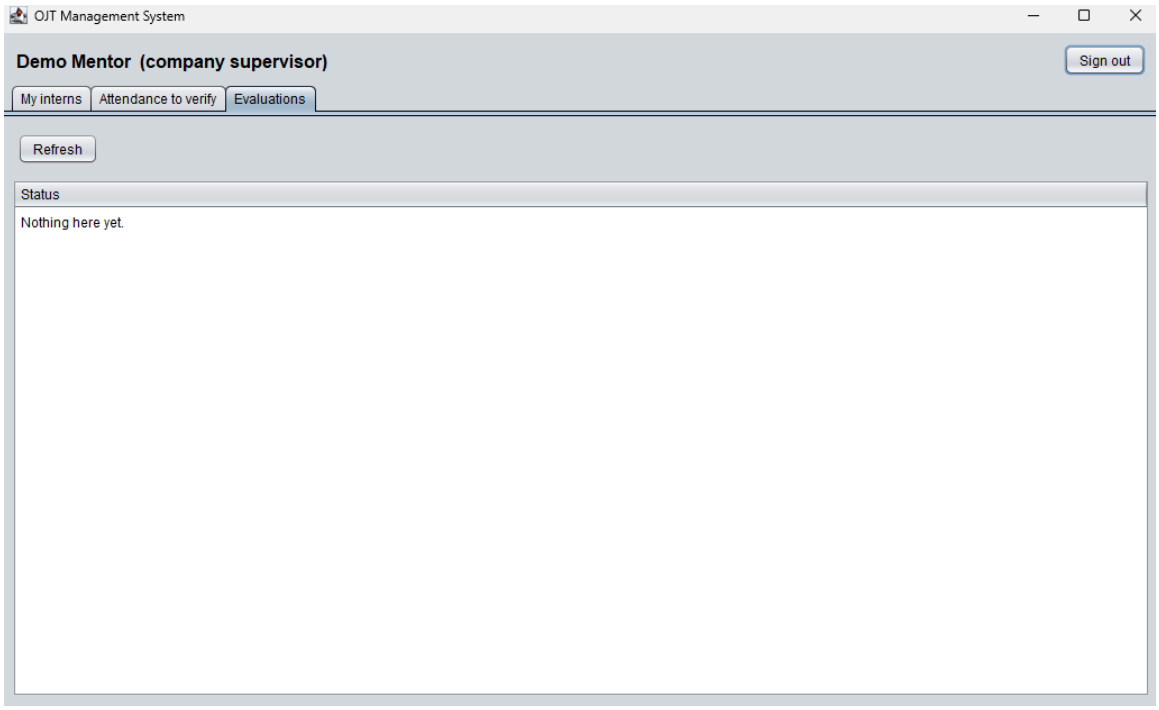
Company supervisor



Company supervisor: My interns.

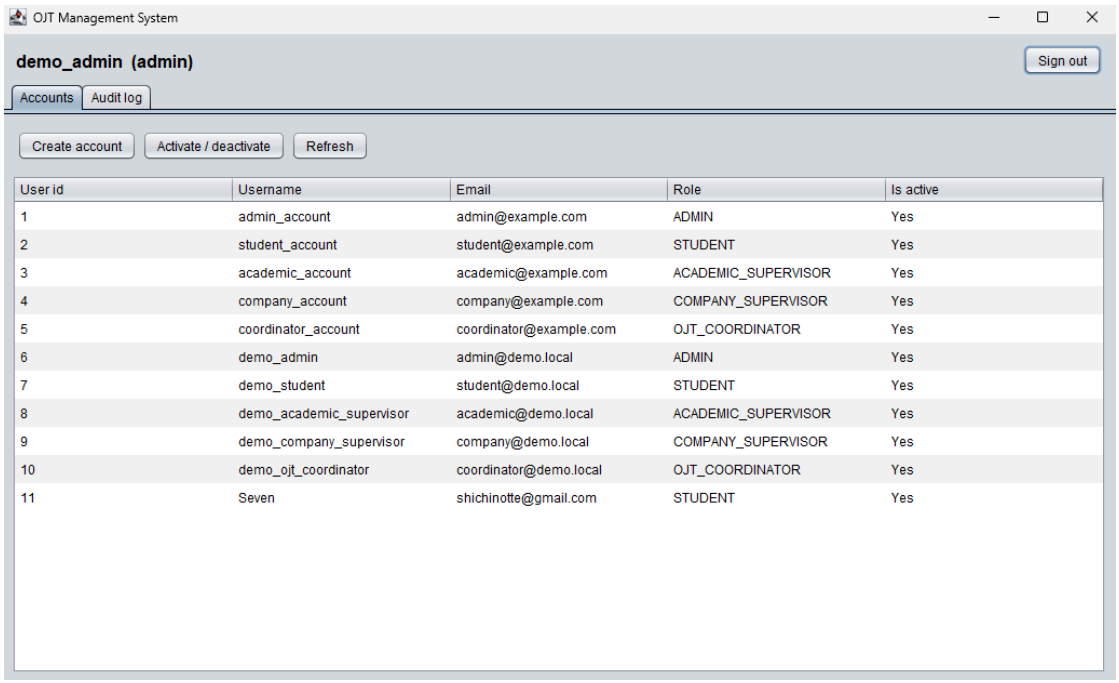


Company supervisor: Attendance to verify.



Company supervisor: Evaluations.

Administrator

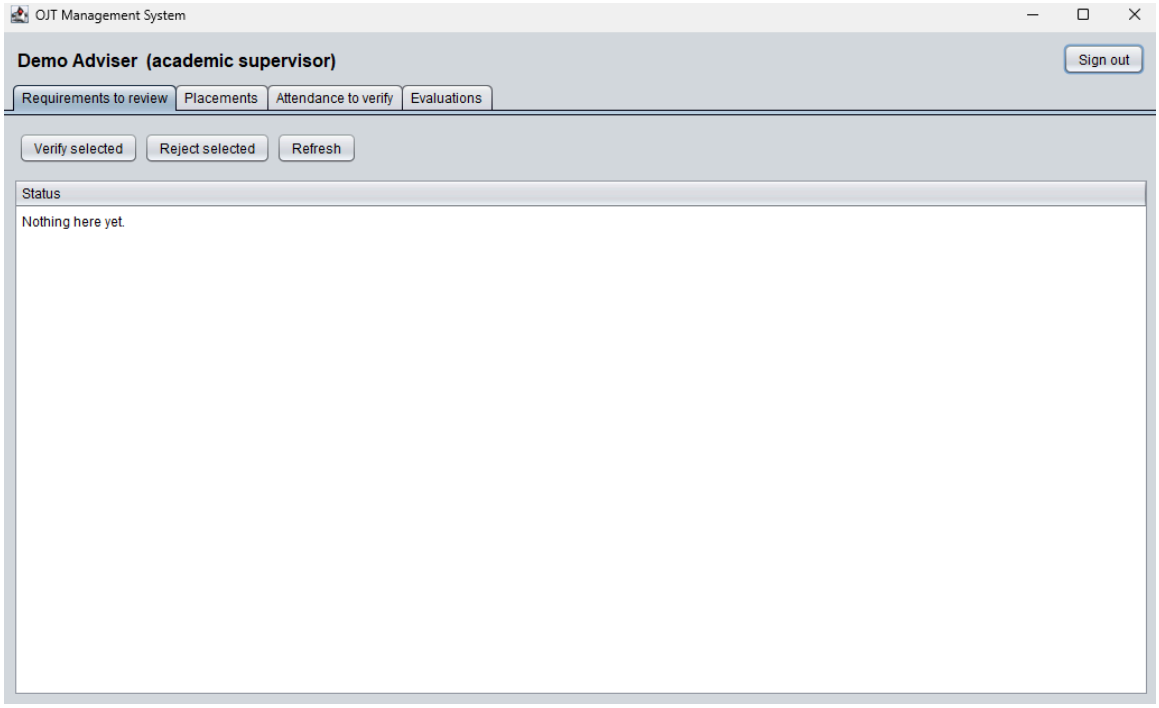


Administrator: Accounts.

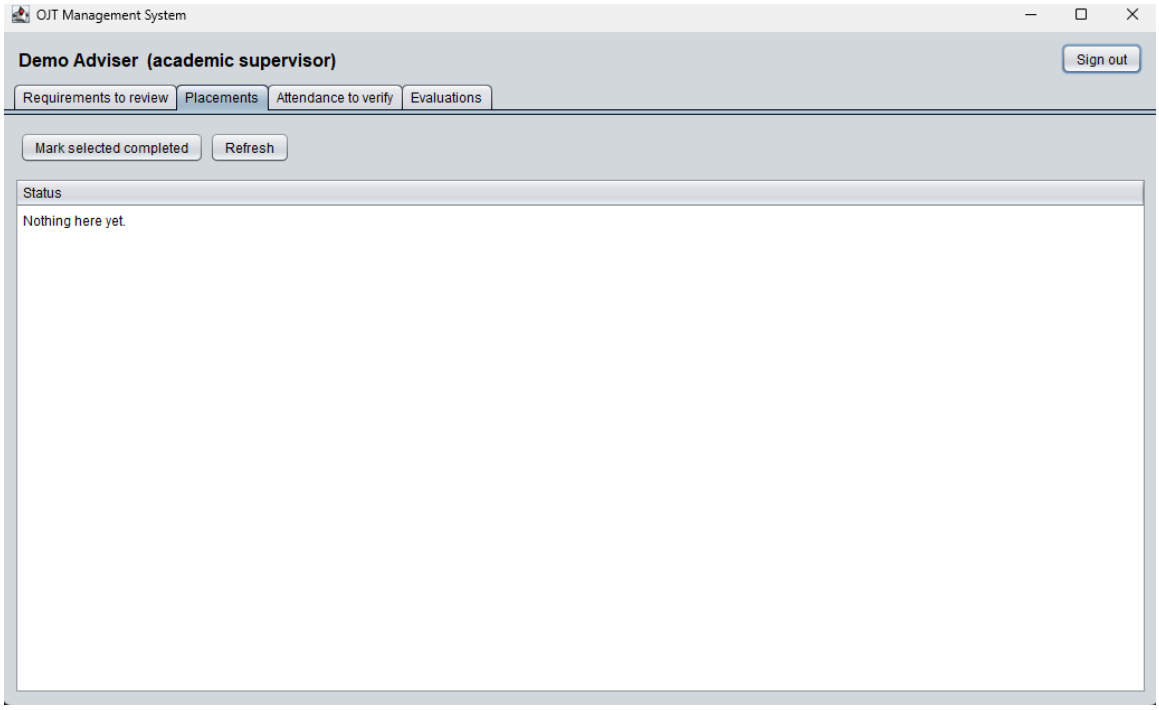


Administrator: Audit log.

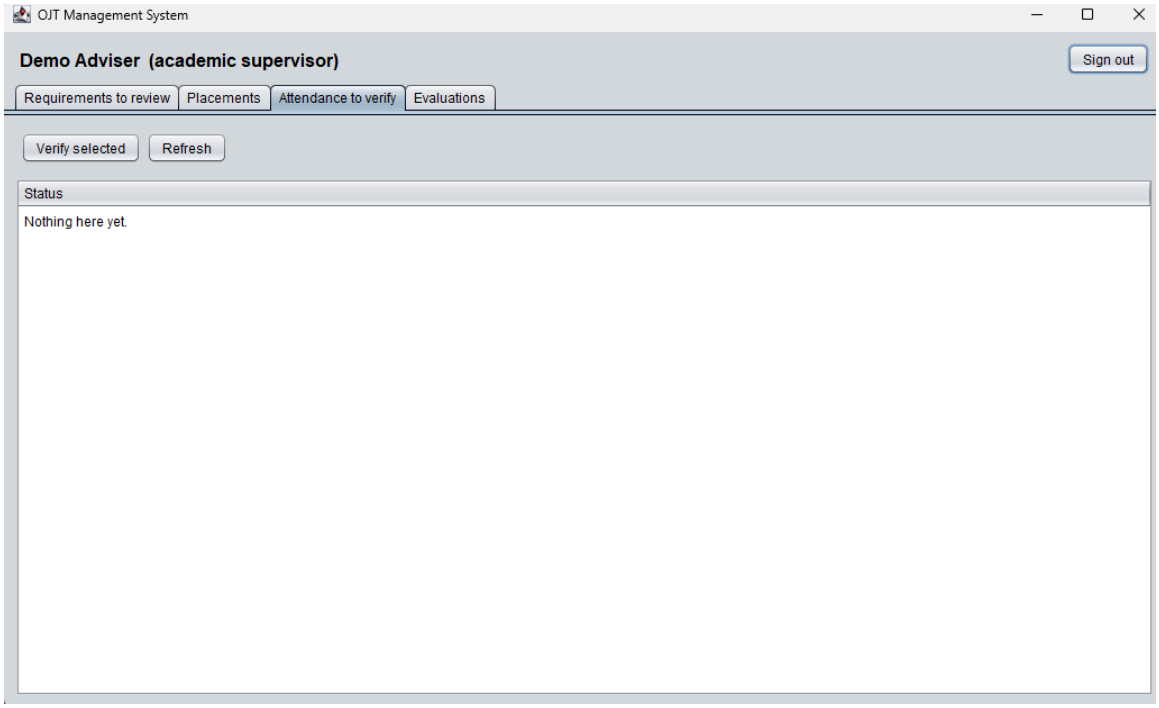
Academic supervisor



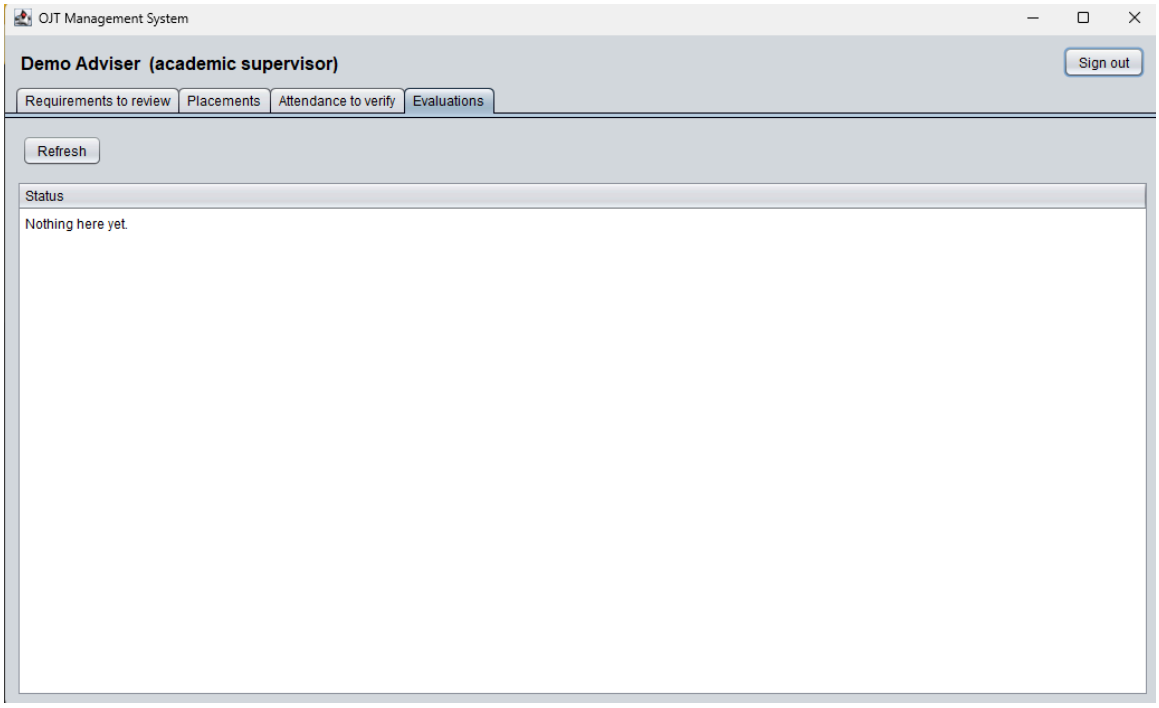
Academic supervisor: Requirements to review.



Academic supervisor: Placements.

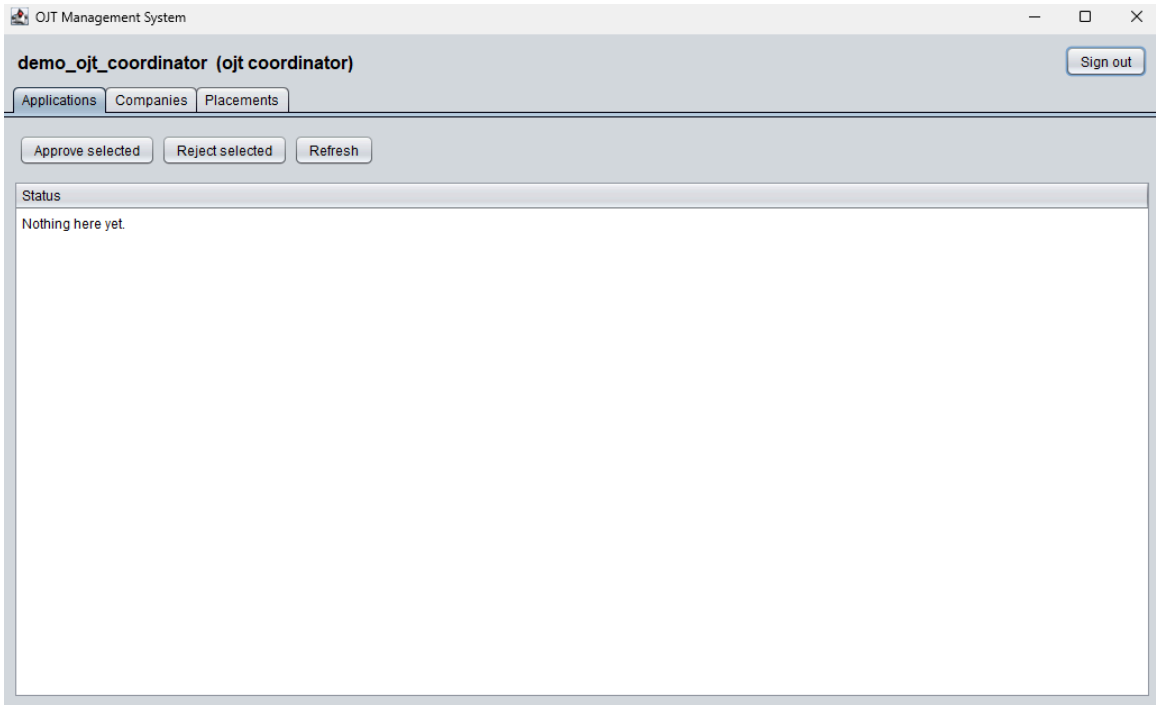


Academic supervisor: Attendance to verify.

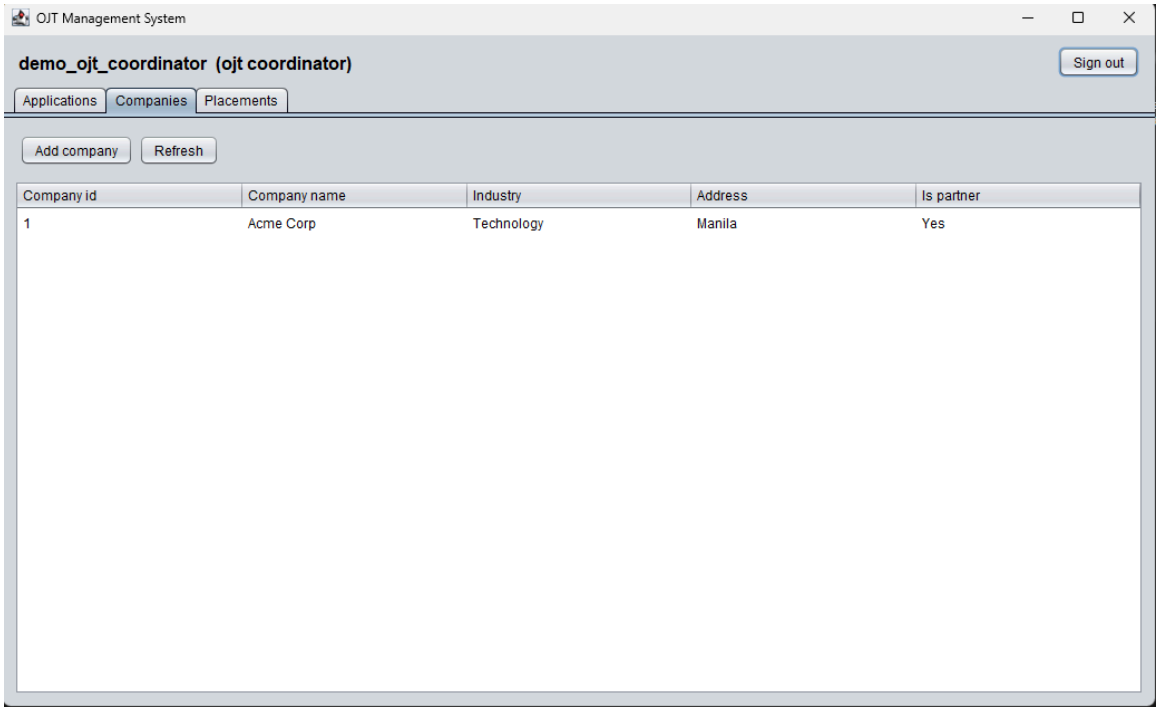


Academic supervisor: Evaluations.

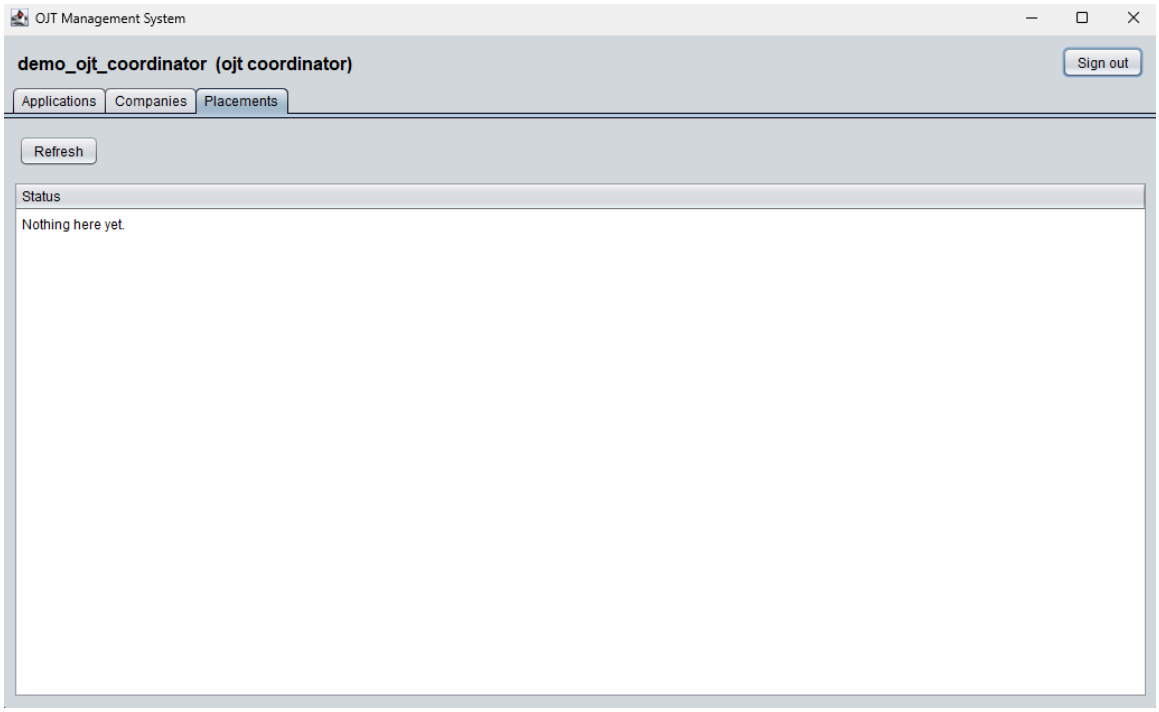
OJT coordinator



OJT coordinator: Applications.



OJT coordinator: Companies.



OJT coordinator: Placements.

CRUD test results

The group ran each test below and all passed as expected. Screenshots cover the Read screens and the account creation; the other results are recorded by the group's own check.

Test	Role	Operation	Actual result
Create an account	Administrator	Create	Account 11 appears in the list
Apply to a company	Student	Create	*Passed*
Submit a requirement	Student	Create / Update	*Passed*
Record attendance twice on one day	Student	Create	*Passed*
Approve an application	OJT coordinator	Update	*Passed*
Evaluate an intern	Company supervisor	Create	*Passed*
Deactivate an account	Administrator	Update	*Passed*

AI usage declaration

AI Tool	Assistance	How the group validated or corrected it
Claude	Produced a rough draft of the OJT application design, used as a basis for the group's own draft.	The group reviewed the draft, then created and justified its own schema and database layout.
Claude	Generated an draft architectural design flow based on prompted information of the group's idea.	A member reviewed it to build their own understanding of the architecture diagram.
ChatGPT	Asked for assistance on how to properly set up Maven for Java.	The group successfully set up Maven